Biologically Plausible Schema Learning

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Abstract

This study explores the integration of connectionist models and biologically grounded hippocampal simulations to enhance our understanding of spatial memory and schema formation in cognitive science. By employing Randy O'Reilly's biologically plausible models from the Computational Cognitive Neuroscience framework, I augmented and evaluated a hippocampal model that simulates learning and memory recall processes. This model leverages the architecture of the hippocampus to perform complex cognitive tasks, such as pattern separation and completion, essential for spatial navigation and memory differentiation. The approach combines the Room Schema attractors, originally conceptualized in the Parallel Distributed Processing volumes, with a more biologically grounded hippocampal architecture, allowing us to investigate how environmental cues and similarities between spatial contexts influence memory interference and schema stability. The experimental setup utilizes varied similarities in room and context vectors to simulate real-world scenarios that challenge the network's memory recall capabilities. The findings reveal that greater environmental resemblance enhances memory confusion, underscoring the importance of distinctiveness in schema formation and retrieval. These insights not only contribute to theoretical discussions on cognitive flexibility and discriminative memory in spatial navigation but may also have practical implications for

designing navigational aids, virtual reality systems, and understanding memory disorders. This study exemplifies how merging traditional connectionist approaches with more advanced, biologically plausible neural simulations can provide deeper insights into the cognitive mechanisms underlying spatial memory and learning.

Keywords: Connectionist Models, Hippocampal Simulations, Spatial Memory, Schema Formation, Computational Cognitive Neuroscience, Pattern Separation, Pattern Completion, Memory Interference, Environmental Cues, Neural Simulations, Cognitive Flexibility, Memory Differentiation, Room Schema Attractors.

Introduction

Think of the traditional home, and ask yourself, what makes up your internal map of the home. How do you delineate what room to sleep in or or eat in, etc... What makes your kitchen a "kitchen". When thinking of the buildings and rooms we interact with everyday, an often nuanced detail of our experience is the formation of schemas. Schemas are units of abstract conceptual formations that interact in multiple ways. In cognitive science schemas refer to structured sets of knowledge about common spaces or environments, that include typical objects and their arrangements within those spaces. We have schemas for many things, restaurants, etc… How do we recognize an abstract concept as fitting a certain schema? Furthermore, how do we recognize a specific room under uncertainty? If the room were dimly lit, or if we only saw a glimpse of one end of a room, how do we categorize the room?

One classic approach to unraveling questions like these is in the form of Connectionist models. These models, often referred to as neural networks, have been a cornerstone in the understanding and modeling of cognitive processes. These models are based on the concept of distributed processing, where information is processed and stored across many different units within a network, rather than in a single node. A fundamental aspect of connectionist models is their use of learning rules, which allow the network to adapt based on input data, thereby modeling the learning processes observed in biological systems.

One prominent application of connectionist models is in the development of attractor networks. Attractor networks are a type of neural network where the system dynamics are designed to converge to a stable pattern, or "attractor," which represents a memory or a cognitive state. These networks are particularly useful for modeling memory, as they can recall entire patterns from noisy or incomplete inputs.

Connectionist neural network models, as crafted in the Parallel Distributed Processing (PDP) volumes, have provided a rich framework for understanding cognitive processes through the lens of distributed representations and the dynamics of network-level interactions. Chapter 14 of the PDP volumes introduces the concept of Room Schema attractors, which are a type of neural network model designed to capture the essence of spatial environments and their associative properties in memory. These models utilize attractor dynamics to stabilize memory representations, allowing the system to remember and reconstruct complex inputs such as the layouts of rooms based on partial or noisy data.

The Room Schema attractor model, serves as an ideal case of how attractor networks can be applied to cognitive tasks such as spatial memory and navigation. The model is designed to encode and recall layouts of rooms, or complex spatial configurations, by stabilizing the memory representation in the form of attractors. This allows the model to fill in missing or degraded information and recover whole representations from partial cues, which mimics the way humans can visualize entire rooms from just a few remembered details, or from a dimly lit scene.

The mechanics of the Room Schema model involve a network of units where each unit is connected to others in a way that reflects the spatial and associative properties of different room layouts. For example, the presence of certain features like a bed or size of the room can be encoded in the connections between units, influencing how information about the room is integrated and recalled. Learning in this model is typically achieved through Hebbian-like rules, where the connections between frequently co-activated units are strengthened, thus reinforcing common patterns or schemas of rooms.

Complementing the PDP approach, Randy O'Reilly's work in "Computational Cognitive Neuroscience" (CCN) introduces biologically plausible models of the hippocampus that incorporate principles of real neural architectures, such as excitatory and inhibitory neural interactions and the use of realistic synaptic learning rules. The hippocampus is crucial for spatial memory and navigation, as evidenced by its role in encoding and retrieving cognitive maps in mammals.(O'Keefe, J., & Nadel, L. (1978))

The CCN textbook provides a comprehensive exploration of neural network models that are biologically realistic, particularly through the use of the Leabra framework. Leabra stands for Local, Error-driven and Associative, Biologically Realistic Algorithm. This framework integrates principles from both error-driven and Hebbian learning and is characterized by its use of both rate-coded and spiking neurons to closely mimic real neural processes. Leabra also incorporates detailed mechanisms of synaptic transmission including excitatory and inhibitory dynamics, which enhance the model's authenticity to actual neural function. These features make the Leabra models particularly effective for simulating detailed brain areas like the hippocampus,

enabling a more nuanced understanding of complex cognitive functions such as memory consolidation.

In contrast to the abstract nature of PDP models, the hippocampal model provides a more biologically grounded approach. The hippocampus is a critical region of the brain involved in the formation of new memories, spatial navigation, and the consolidation of information from short-term to long-term memory. (Eichenbaum, H. (2000)). The primary goal of the current project is to revamp the connectionist neural network attractor model, originally discussed in the context of Room Schema attractors, using a more biologically plausible hippocampal model from the CCN framework. By integrating these two approaches, the project aims to achieve a more realistic simulation of how spatial schemas are formed, maintained, and utilized in the brain.

Methods

The model used in the current study is from Randy O'Reilly's neural network simulations for the Computational Cognitive Neuroscience (CCN) Textbook. The "hip" model used is from chapter 9 and leverages a hippocampal architecture that simulates the dynamic processes of learning and memory recall found in biological systems. It's organized into several key areas including the input layer, ECin, DG, CA3, CA1, and ECout layers, which are essential for encoding, pattern separation, pattern completion, and memory recall. This structure not only reflects the complex organization of the hippocampus but also its functional dynamics.

The model's neural activation flow begins at the Input Layer, which receives external inputs for memory encoding, representing the interface with the neocortex via the entorhinal cortex. This layer projects to the ECin (Entorhinal Cortex input) Layer, facilitating both inbound and outbound communication with the hippocampus. The ECin layer then connects to the Dentate Gyrus (DG) Layer, which performs pattern separation to distinguish similar input patterns and reduce memory recall interference. Subsequently, the DG layer sends projections to the CA3 layer, involved in pattern completion by recalling memories from incomplete cues and maintaining autoassociative memory functions essential for interference-free processing. The CA3 layer links to the CA1 layer, the hippocampus's output region, which forms the final memory representation sent back to the entorhinal cortex. Finally, the CA1 layer connects to the ECout Layer, completing the feedback loop to the entorhinal cortex and distributing processed information to other cortical areas.

This model incorporates the theta phase to organize error-driven learning across the EC to CA1 projections and other connections. This approach aligns synaptic changes with the biologically realistic oscillating rhythms in the brain, believed to be critical in learning and memory.(Buzsáki, G. (2002).)After training, the network undergoes testing phases where it needs to recall the learned patterns based on partial inputs. This phase is crucial for evaluating the effectiveness of the memory model and its resilience against interference—how well it can differentiate between similar memories when cues are similar but not identical.

In my study, I utilized the AB/AC list learning paradigm, a paired associates learning task widely used to understand memory and interference. Methodologically, this task involves presenting subjects with pairs of associated items (e.g., vector pairs) in a sequence to be learned,

referred to as the AB phase. Subsequently, one item of some pairs is re-paired with a new item in what is termed the AC phase. The critical challenge here is to learn the new AC pairings without interfering with the memory of the original AB pairings. The methodology taps into the neural and cognitive mechanisms responsible for managing interference and increasing recall precision, making it relevant for studying how memories are differentiated and retrieved under overlapping conditions.

In this current simulation, this paradigm was adapted to investigate spatial memory by using types of rooms and specific attributes as associative pairs. The "A" input represented a generalized room type, like a kitchen, while the "B" input and "C" input were specific features or attributes of rooms within different houses. Simply, “B” is a singular kitchen in the “B house”, and “C” is another kitchen, but in the “C house”. After training the network to associate these pairs in the AB configuration, I introduced the AC pairs, where "A" remained the same but "C" represented a different attribute of the same room type in another house. This setup allowed me to explore how environmental and architectural elements are encoded and recalled, particularly focusing on how similar or overlapping features influence memory recall and interference in spatial contexts. The test phases involved presenting partial inputs (just the "A" and "context" vectors) and observing the network’s ability to accurately recall the associated "B" or "C" features, which evaluates the model’s resilience against interference and its capacity for accurate memory retrieval.

The "context" vectors were specifically used to represent two different houses, with each house providing a unique background setting that influences the association between the "A" inputs (room types) and the "B" or "C" inputs (specific room attributes). This design allowed for

the examination of how the architectural context, as differentiated by the two houses, impacts the memory encoding and retrieval processes, shedding light on how environmental distinctions can either mitigate or exacerbate memory interference.

Utilizing the feature set from the original 'Room Schema model' of PDP, I developed a layout for the hippocampal model's 'B' pattern activations that corresponds to specific room features, as depicted in Fig. 1. Five distinct rooms were created—Office, Kitchen, Living Room, Bedroom, and Bathroom—each containing a unique input pattern comprising subcategories 'A', 'B/C', and 'Context'. These rooms were aligned with the categorizations used in the original connectionist model. The patterns were sequentially presented to the model in the order of; a*_0, a*_1 , a*_2, a*_3, a*_4 (Where * is either B or C list blocks).

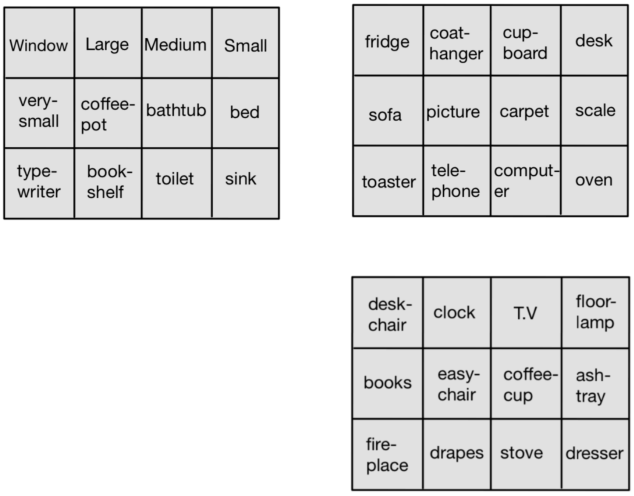


Figure 1: Mapping of Room Features to a Pattern Grid for Hippocampal Model Simulation: This figure illustrates the structured layout of various room features categorized by size and features, adapted from the Room Schema model of PDP to fit the hippocampal simulation model from CCN. The grid organizes objects to fit in the “B” or “C” patterns. Each cell represents a potential

neural unit activation in the hippocampal model, facilitating the study of spatial memory and object association within architectural contexts.

To delve deeper into the dynamics of memory interference, I modified the encoding of the input vectors. By adjusting the similarity between the context vectors for the AB and AC configurations, I simulated the effects of the houses being more similar in appearance or structural layout. This modification allowed us to explore several questions regarding the impact of environmental and decorative similarities on memory interference. For instance, I found that an increased similarity between the kitchens of two houses heightened memory interference. Additionally, when many rooms shared identical features across both houses, this significantly challenged the model's ability to differentiate the unique features of the rooms that differed.

Through this methodological approach, this study systematically manipulated the activation patterns representing various architectural features and spatial contexts, allowing for a comprehensive exploration of how architectural and contextual similarities between environments affect spatial memory and associative learning in the hippocampal model.

Results

Initially, the model is trained on the set of predefined room patterns, which are part of the AB training set. Each training pattern is composed of three parts: 'A' (the prototype across all rooms), 'B' (specific attributes unique to each room in the first scenario), and 'Context' (additional contextual cues). During the training, these patterns are presented to the network in the aforementioned structured manner, where activation flows sequentially from the ECin layer,

through the pattern-separation process in Dentate Gyrus, to the CA3 region for pattern completion, and finally to CA1.

After training, the model goes through a testing phase and attempts to recall the learned patterns. This phase is crucial in evaluating the learning process. During testing, the model is presented with partial cues, missing the 'B' component, which it must complete based on the learned associations. The sequences of patterns during this phase include both AB (initial learning set) and AC (new learning set introduced to examine the model's ability to adapt to new information without interfering with previously learned information). The testing also introduces 'Lure' trials, which contain completely novel patterns to assess the model's response to unfamiliar stimuli.

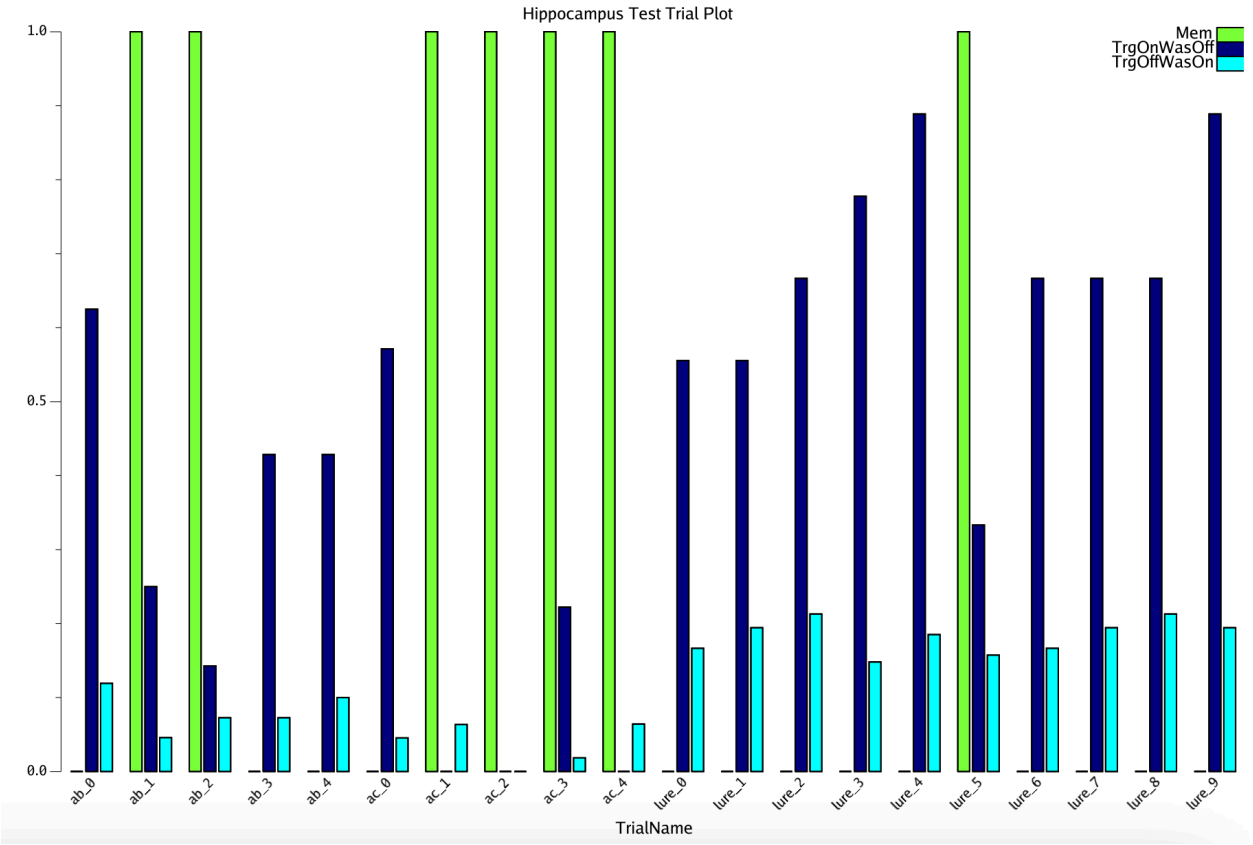


Figure 2: This bar graph shows the outcomes of individual test trials for different stimulus sets: AB, AC, and Lure. Each set of bars represents a different trial, and the trials are labeled on the x-axis (e.g., ab0, ab1, ac0, lure0, etc.).

In the trials, I observed distinct patterns of memory recall accuracy, which were measured and categorized into three primary types: AB, AC, and Lure trials. Each category was evaluated based on Memory Accuracy (Mem), with perfect recall represented by a score of 1.0, and two types of errors: TrgOnWasOff, where necessary memory units failed to activate, and TrgOffWasOn, where unnecessary units were incorrectly activated.

The results from the AB trials showed consistently high memory accuracy, with most trials achieving scores near or at 1.0. This level of recall accuracy displays the model's capability

to effectively encode and retrieve memories when the learning and recall conditions are stable and unchanged. The green bars (Mem) are almost consistently high across all AB trials, showing excellent recall accuracy. This suggests that the network has effectively learned and is able to recall these patterns with high fidelity. Both error types (navy and cyan) are generally low in these trials, indicating minimal confusion or misactivation.

Mem Accuracy in the AC trials are high but generally show a slight decrease compared to AB trials, suggesting some challenges in recall accuracy, likely due to interference or confusion with similar AB memories. The navy bars show increased errors, indicating more frequent failures in activating required units, while cyan bars also show errors but are not as prominent.

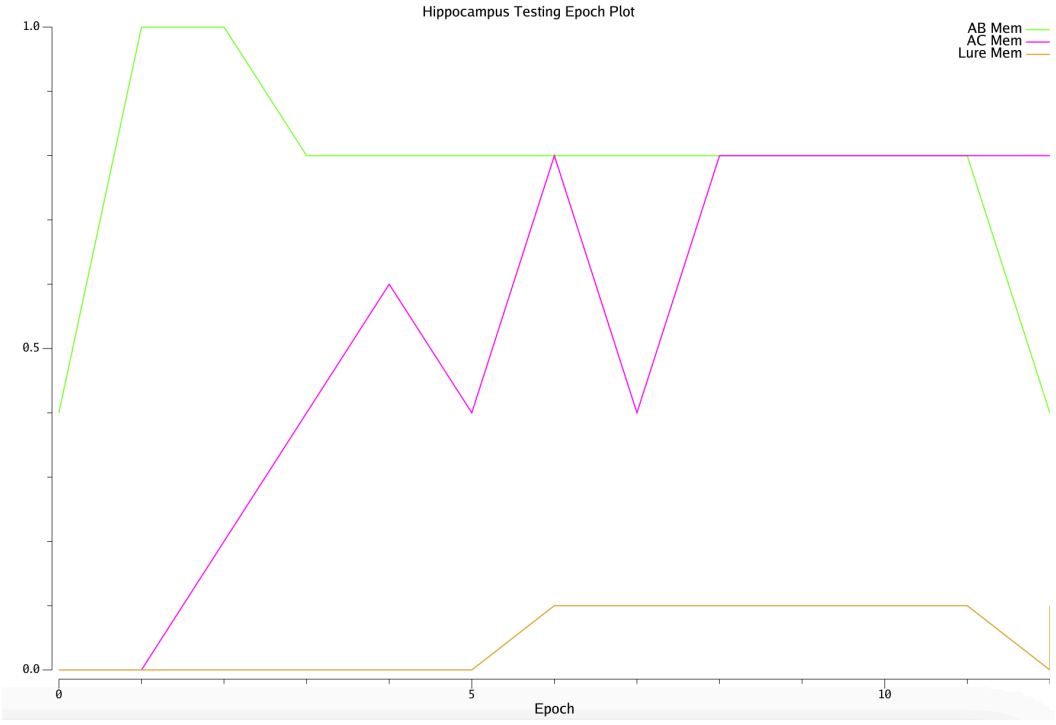


Figure 3: This line graph tracks memory performance across multiple epochs of testing.

Each line in Fig.3 represents the memory recall accuracy on the initial test over time for the different sets of patterns: The green line represents the AB memory recall, and remains consistently high throughout the testing epochs. This indicates that once the AB patterns are learned, the network retains and recalls this information across subsequent epochs. The stability of the AB line suggests strong initial learning and effective long-term memory retention for these patterns.

The pink line shows AC memory recall, and exhibits more variability compared to the AB line. It shows fluctuations, with dips and recoveries. This pattern reflects the network's ongoing process of integrating new AC memories while simultaneously managing interference from the previously learned AB patterns. The dips may represent moments of confusion or conflict between similar but distinct memories, with recoveries indicating successful adaptation and re-stabilization of memory recall after adjusting to the new information.

The orange line, which tracks the recall of Lure patterns, shows the most significant fluctuations. This line generally scores lower than AB and AC, reflecting the difficulty the network faces when dealing with entirely novel and non-trained stimuli. The variability in this line, by dips and recoveries, suggests a trial-and-error process in handling these unfamiliar inputs. Each dip may represent a failure or challenge in recalling these new stimuli, while each recovery points to partial successes or adjustments in the network's approach to processing novel information.

Overall, the hippocampal model effectively learns and recalls AB and AC patterns with high accuracy, demonstrating its ability to handle overlapping and distinct memory patterns. The fluctuations in AC and Lure memories across epochs could suggest dynamic adjustments within the network, possibly tweaking its internal representations to optimize both old and new information. The results validate the network's design for tasks requiring memory differentiation and pattern completion, crucial for tasks like spatial navigation and learning in dynamic environments.

To perturb the model, In an extension of my hippocampal simulation study, I examined the impact of varying degrees of similarity on memory interference by manipulating the resemblance between room ("B"/"C") and context ("houses") patterns. This approach allowed me to explore how subtle changes in environmental cues affect the neural network's ability to recall spatial configurations under conditions that closely mimic real-world scenarios of memory confusion, such as mistaking one location for another due to their similarities.

I manipulated the similarity between the "B" and "C" patterns to reflect varying degrees of resemblance between specific rooms of different houses: Low Similarity (33%): Rooms shared minimal features, such as one or two pieces of furniture. No shared units/1 shared unit. Moderate Similarity (50%): Rooms shared more features, including furniture types and layout but differed by 2 units. High Similarity (75%): Rooms were almost identical in layout and major furniture items but had 1 different unit. Similarly, I manipulated the similarity in the "context" vectors to reflect different architectural styles and structural layouts of the houses Low Similarity (25%): Houses shared basic structural frameworks but differed markedly in design, such as a ranch-style versus a multi-story house.No shared units/1 shared unit. High Similarity (75%):

Houses shared many structural elements, such as the same number of rooms and similar floor plans, but featured different exterior designs and landscaping.1 differing unit.

Initial results indicated a significant impact of similarity on memory performance. In the low similarity condition (33% for rooms and 25% for houses), the network demonstrated robust differentiation skills, with high recall accuracy in both AB and AC trials and minimal interference. As the similarity increased to 50% and 75% for rooms, I observed a progressive decline in the network's ability to distinguish between the AB and AC memories, particularly when the context similarity was also high (75%). This was evidenced by a decrease in recall accuracy for the AC trials and an increase in memory confusion errors, where the network would often recall attributes of an AB room when tested with AC cues, as shown in Fig. 4 with high similarity and Fig.5 with moderate similarity.

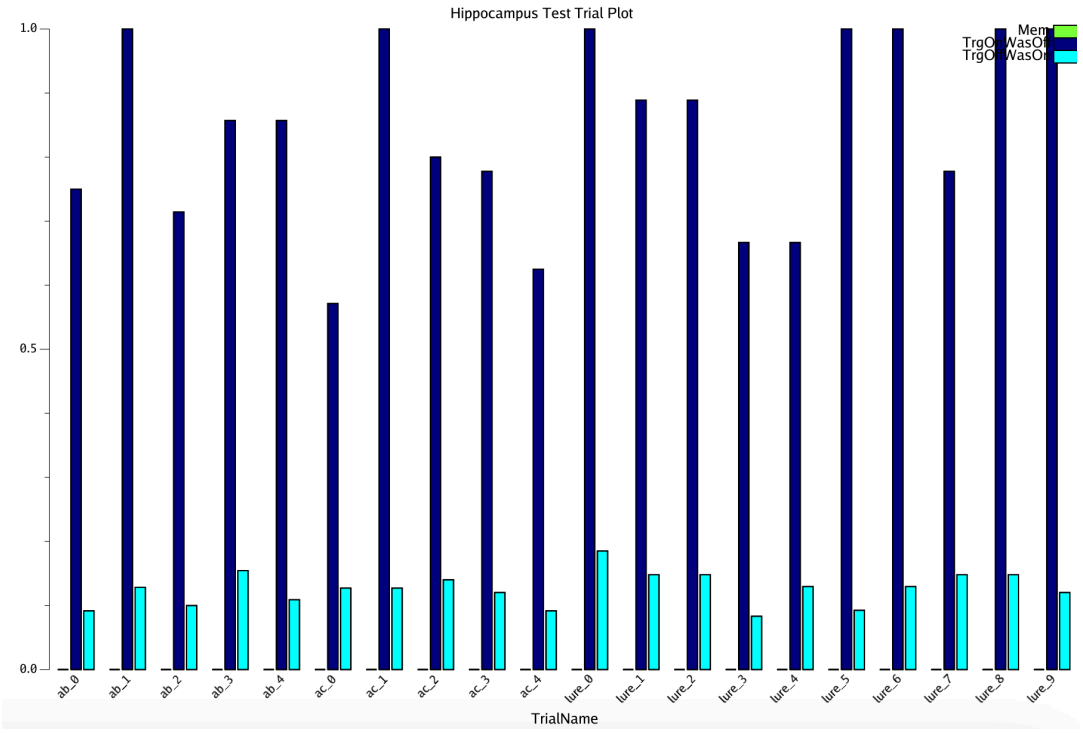


Figure 4: High similarity(75%) context and room condition. The navy blue bars indicate the proportion of correct recalls (Mem), with a height of 1.0 representing perfect memory recall. The cyan bars represent memory confusion errors (TrgOffWasOn), where the network incorrectly activates units that should have remained inactive. As the similarity between room features increases, a notable decrease in recall accuracy for AC trials can be observed, particularly when contextual similarity is also heightened. This graph visualizes the impact of increasing similarity on memory performance, confirming that higher resemblance levels lead to increased memory confusion and reduced recall accuracy.

From Fig.4, it's evident that there are significant recall challenges across both AB and AC trials, especially with the high incidence of TrgOnWasOff errors (navy bars). This indicates that while the network might not be confabulating excessively (as seen by relatively lower cyan bars in many trials), it is failing to activate significant parts of the correct patterns. This type of error suggests issues with pattern completion and recall effectiveness, potentially increased by the similarity manipulations in the room and context patterns.

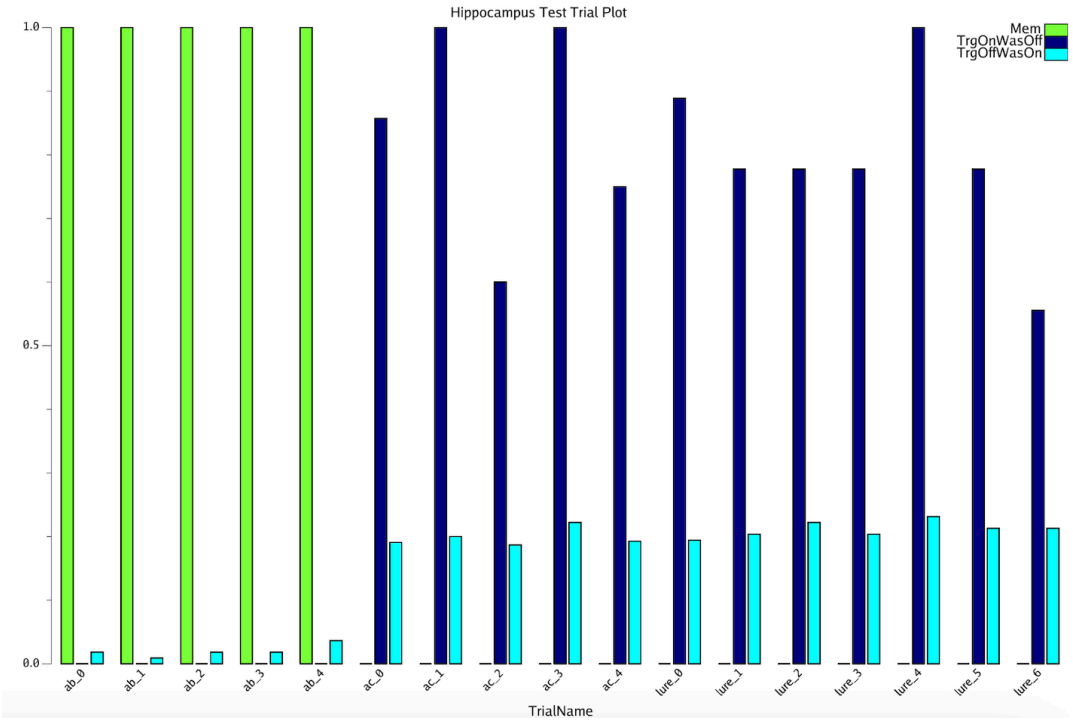


Figure 5: The AC trials with Moderate Similarity (50%) show some memory confusion, likely due to the overlap in features making differentiation harder.

While some trials maintain high recall accuracy, there is an evident increase in both types of errors, particularly the TrgOnWasOff errors (navy bars), which suggests some difficulty in pattern differentiation due to the moderate increased similarity.

Discussion

The increase in similarity between the rooms led to substantial interference in the network's memory recall capabilities, supporting the hypothesis that greater environmental resemblance enhances memory confusion. This effect was amplified by high context similarity, suggesting that both specific attributes of rooms and their broader contexts play crucial roles in how memories are encoded and recalled. These findings align with theories of pattern separation

and completion in the hippocampus, which suggest that distinctiveness of cues is critical for reducing interference and enhancing memory accuracy.

The study provides valuable insights into the neural dynamics of spatial memory and the conditions that may exacerbate or mitigate memory interference. By systematically varying the similarity in environmental cues, this demonstrated how neural networks might simulate the interactions between different memory systems and the challenges faced in real-world memory applications, such as navigating similar environments or distinguishing between locations with subtle differences.

The trials involving varied similarities of rooms and contexts, directly relates to and extends the foundational ideas of the Room Schema model in several significant ways: The simulation experiments with varying levels of similarity between rooms and their contexts enhancing understanding of how environmental details are integrated into room schemas. The original Room Schema model posits that these schemas are not just static representations but dynamic constructs that adapt based on input. By manipulating the similarity of these inputs, this study provides insights into how changes in perceived environment can influence the stability and accuracy of these schemas, directly relating to how humans might navigate and remember real-world spaces.

One of the core challenges addressed in this study is the issue of memory interference, a topic that is also central to the Room Schema model. In real-world terms, this would be akin to the difficulty one might experience in distinguishing between two similar hotel rooms visited in quick succession. The current model's focus on the effects of similarity between different but closely related schemas (rooms and houses) illuminates how the brain might handle such

interference, contributing to theoretical discussions on cognitive flexibility and the capacity for discriminative memory in spatial navigation.

The findings from the hippocampal model simulation provide empirical support for several theoretical aspects of the Room Schema model. For one, the degradation in performance with increased similarity supports the schema theory's assertion that distinctiveness is key to schema formation and retrieval. It also highlights the importance of context in schema activation, suggesting that similar contexts can lead to schema blending or confusion, which is a critical consideration for both cognitive modeling and real-world applications like architectural design or virtual reality environments.Ultimately, the relationship between this hippocampal simulation work and the Room Schema model from PDP provides a robust framework for applications beyond basic research. For example, it could influence the design of navigational aids, enhance virtual reality systems for training or therapy, and improve our understanding of memory disorders by identifying how schema disruption might occur in conditions like Alzheimer's disease.

This study emphasizes the potential of integrating connectionist models with biologically grounded hippocampal simulations to enrich our understanding of spatial memory and schema formation. By employing Randy O'Reilly's advanced neural architectures within the Computational Cognitive Neuroscience framework, the model demonstrated how such integrations can not only mimic but also deepen our comprehension of complex cognitive processes. The findings reveal that environmental similarities significantly influence memory interference, highlighting the critical role of distinctiveness in effective schema formation and

retrieval. This research not only confirms theoretical discussions in cognitive science but also holds promise for practical applications in developing navigational aids, virtual reality systems, and interventions for memory disorders. Moving forward, the continued fusion of connectionist approaches with biologically informed models stands to offer invaluable insights into the neural underpinnings of memory, and may potentially revolutionize strategies for cognitive enhancement and rehabilitation.

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Appendices

Appendix A: Simulation Software and Code

Description:

Software Used is Leabra, developed by Randall O'Reilly et al., implemented through the Emergent software package, which provides a platform for creating and testing complex models of brain function based on the Leabra algorithm.

The simulations were performed using Emergent version 9.0.1.

Installation Instructions:

Ensure your system meets the software requirements: Windows 10, macOS 10.14+, or a recent Linux distribution.

Download the version of Emergent from the GitHub repository: Emergent GitHub

Follow the detailed installation instructions provided in the repository to set up Emergent on your operating system.

Accessing the Source Code:

Visit the GitHub page: Emergent on GitHub Clone the repository to your local machine using the command: git clone https://github.com/emer/leabra.git

The source code can be navigated and modified as required for specific simulations. Comprehensive documentation is available on the GitHub wiki: Emergent Documentation

Contents:

GitHub - CompCogNeuro/sims: simulations for the Computational Cognitive Neuroscience textbook

sims/ch8/hip at master · CompCogNeuro/sims · GitHub

Appendix B: Data Files

Description: Original Room patterns hard coded in binary csv

Contents: https://github.com/mgilbert2/Testing-Training-ROOM.git

Appendix C: Methodological Steps for Replication

Description: ‘hip’ model at terminal

Contents:

1. Under TRAINAB-TESTAC in the toolbar,
2. Open each respective activation pattern and use the “open csv” button and upload each respective file provided from appendix b.
3. Init, and train for 30 seconds
4. No manipulations for my first “Test all”
5. Reopen activation patterns for Train and test and set activity parameters to similarity levels described below;

Low Similarity (33%):

Description: involving only one active node within the pattern.

- This was achieved by approximately 33% of the nodes in the "B" pattern were shared with the "C" pattern, with the remaining nodes being unique to each pattern.

Moderate Similarity (50%):

Description: Rooms differing by two nodes.

- Half of the nodes in the "B" pattern were identical to those in the "C" pattern, with two specific nodes actively differing.

High Similarity (75%):

Description: Rooms were almost identical but included one distinctively different node.

- Aligning 75% of the nodes between the "B" and "C" patterns, with only one node actively different.